Systematically missing data in distributed data networks: multiple imputation when data cannot be pooled

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December 12, 2024

International Biometric Conference, Atlanta, GA, USA

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Disclaimer

I have no current or past relationships with commercial entities.

100% missing confounder in two studies

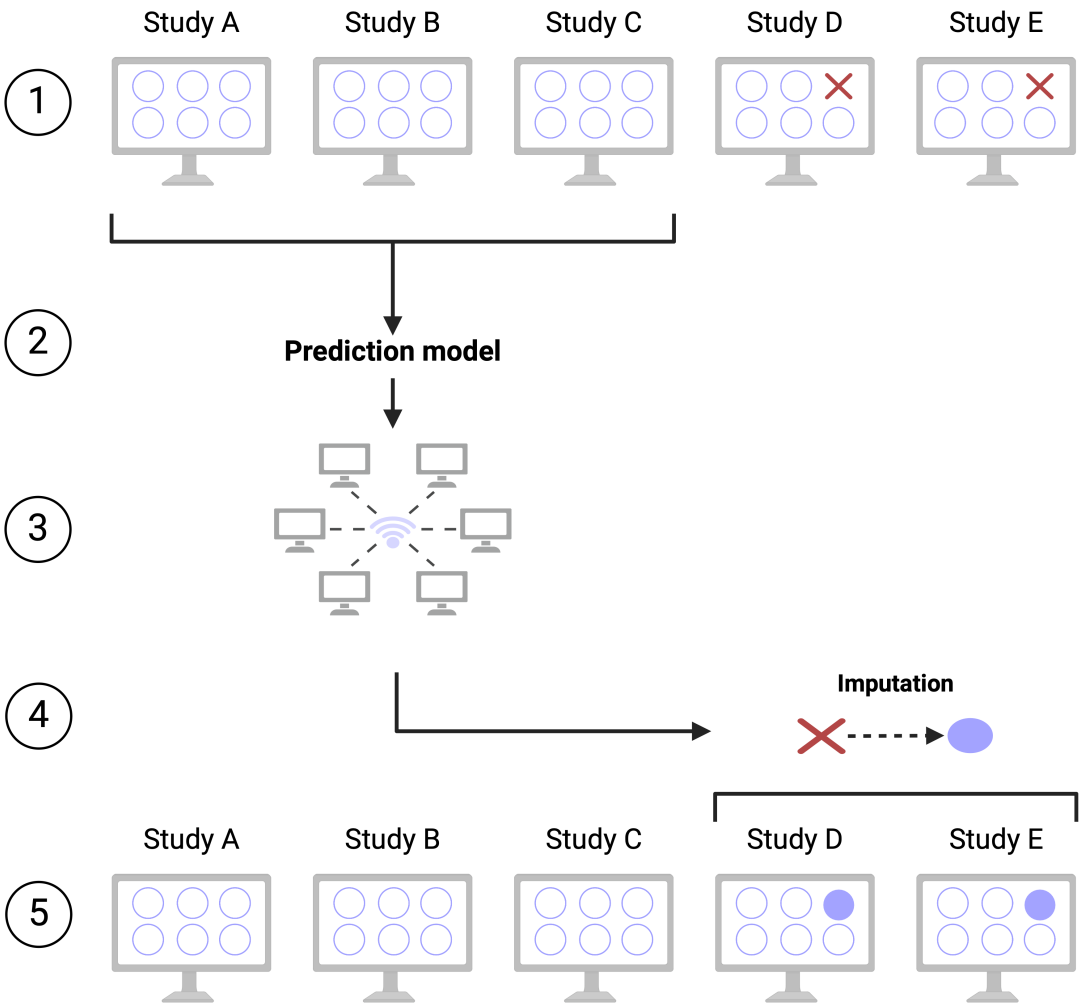
	Study A N=4,182	**Study B** N=2,260	**Study C** N=1,401	**Study D** N=3,766	**Study E** N=2,382
Exp.(%)	23	26	30	15	17
Out.(%)	33	29	28	25	33
paOR	1.8(1.5-2.2)	1.6(1.2-2.0)	1.5(1.3-1.7)	2.0(1.5-2.5)	1.8(1.5-2.1)
faOR	1.3(1.1-1.6)	1.1(0.8-1.5)	1.1(0.9-1.2)	**NA**	**NA**

Missing data in distributed research networks

How do we impute missing data from external data sources?

- Some studies may have not collected some variables due to financial issues, human resources etc.
- Increasingly, individual data *cannot* be shared due to legal or logistic barriers.
- When individual-level data *cannot* be shared between sites, common multiple imputation strategies fail to work due to the lack of basis for imputation.

Imputation using external data



Software available to implement these steps for binary, categorical, and continuous variables.

Imputation method

Consider a continuous variable z_i which is 100% missing in study D.

- In Study A, estimate p-quantile regression model for the continuous variable z_i conditionally on a set of predictors $\mathbf{w}_i$:

$$\hat{Q}_{z_i|\mathbf{w}_i}(p) = \mathbf{w}_i \mathbf{f}(p) \quad p \in \{0.01, 0.02, \ldots, 0.99\} \tag{1}$$

- Export imputation regression coefficients and their variances to be sent back to Study D.
- Optionally, take a weighted average of all imputation regression coefficients from multiple studies.

Assignment of values to the missing data

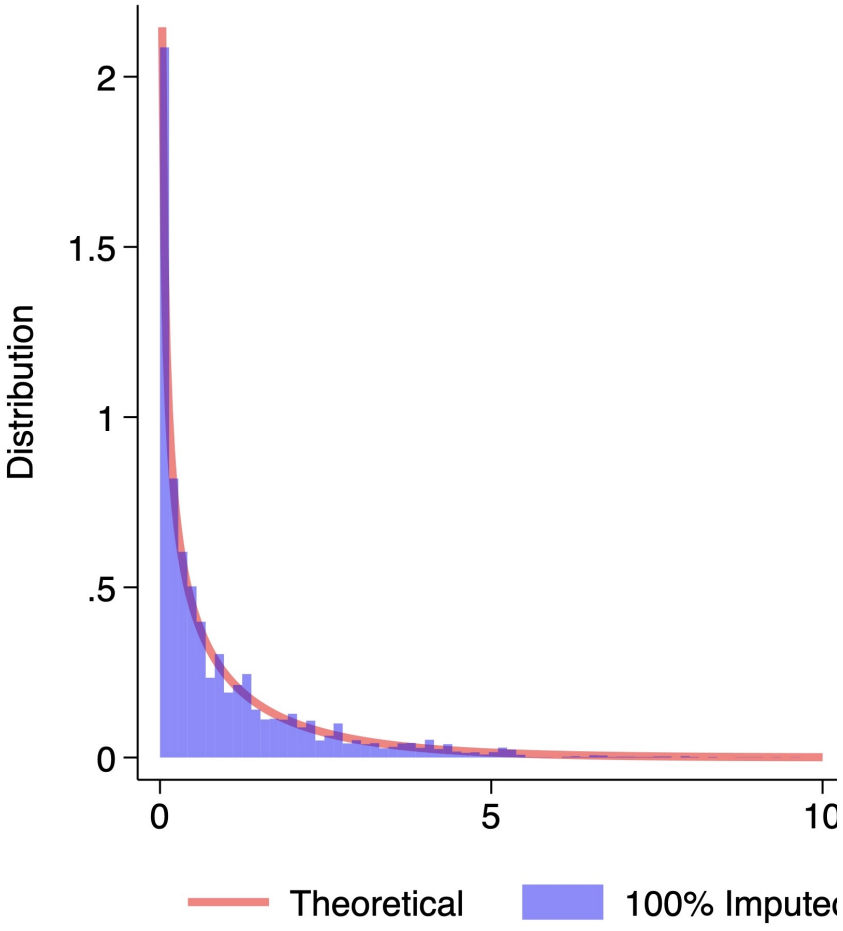
We denote $z_i^{(m)}$ as the m-th imputation of a missing value in z_i. In study D:

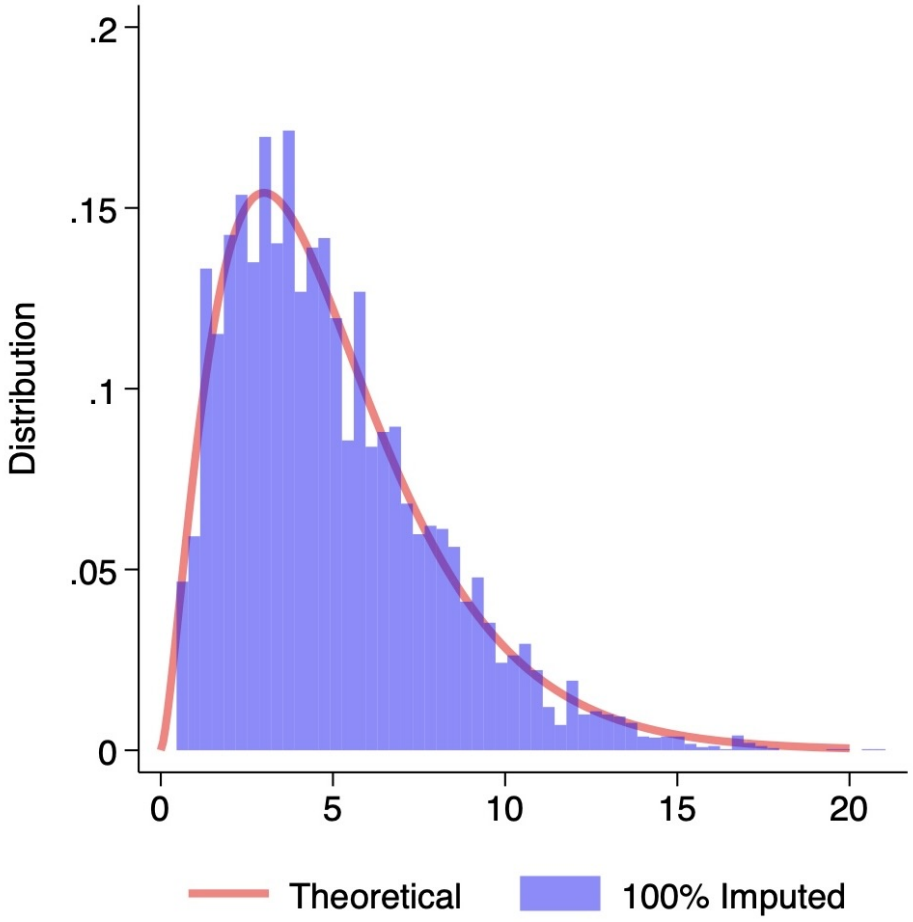
1. Draw a random value U_i from a continuous uniform distribution $\mathcal{U}(0,1)$.
2. Compute the weighted average of the F and $F+1$ conditional predicted quantiles

$$z_i^{(m)} = (1 - \text{mod}) \cdot \hat{Q}_{z_i|\mathbf{w}_i}(F) + \text{mod} \cdot \hat{Q}_{z_i|\mathbf{w}_i}(F+1) \quad (2)$$

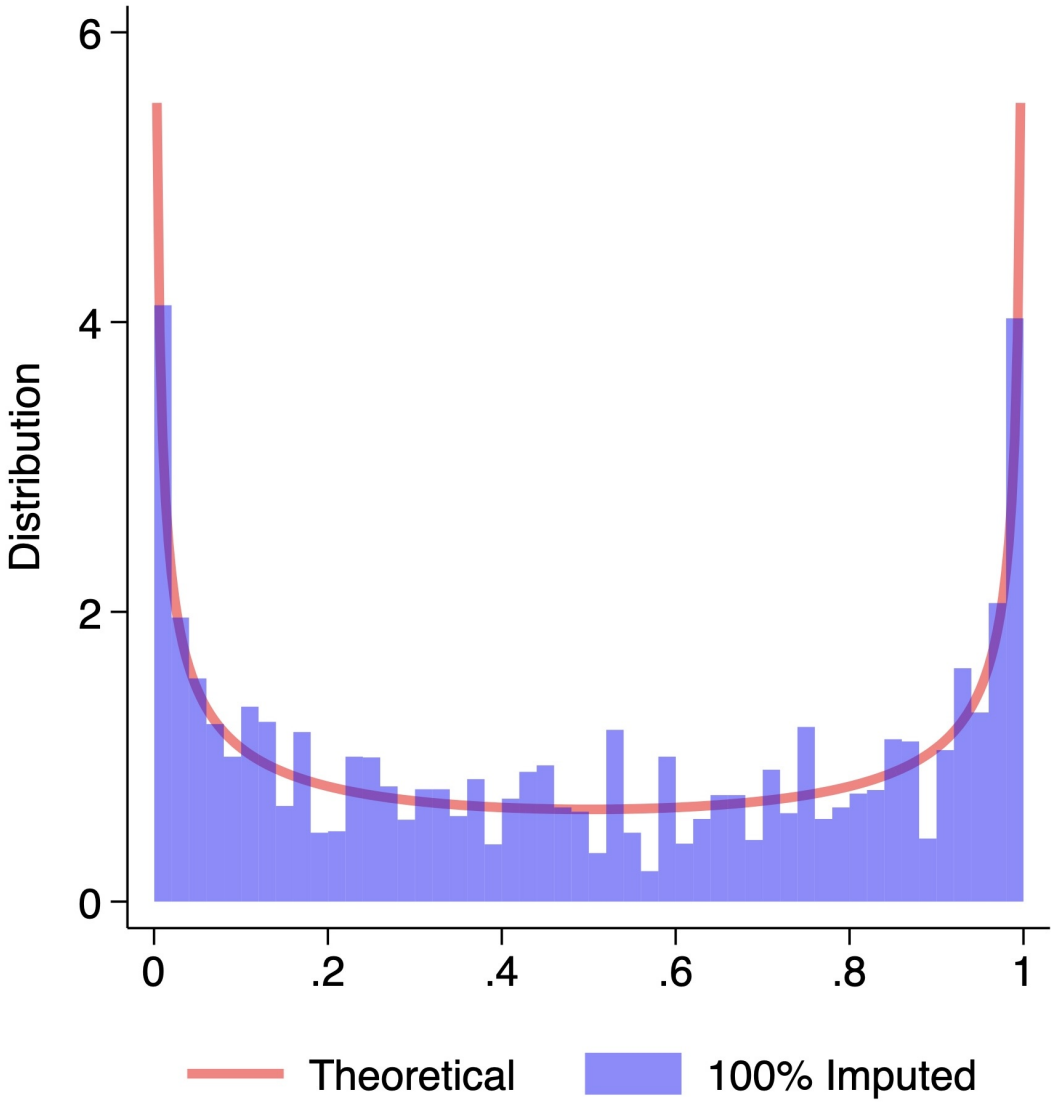
where $F = \lfloor U_i\% \rfloor$ and $\text{mod} = U_i\% - \lfloor U\% \rfloor$.

χ^2-distribution with 1 and 5 degree(s) of freedom





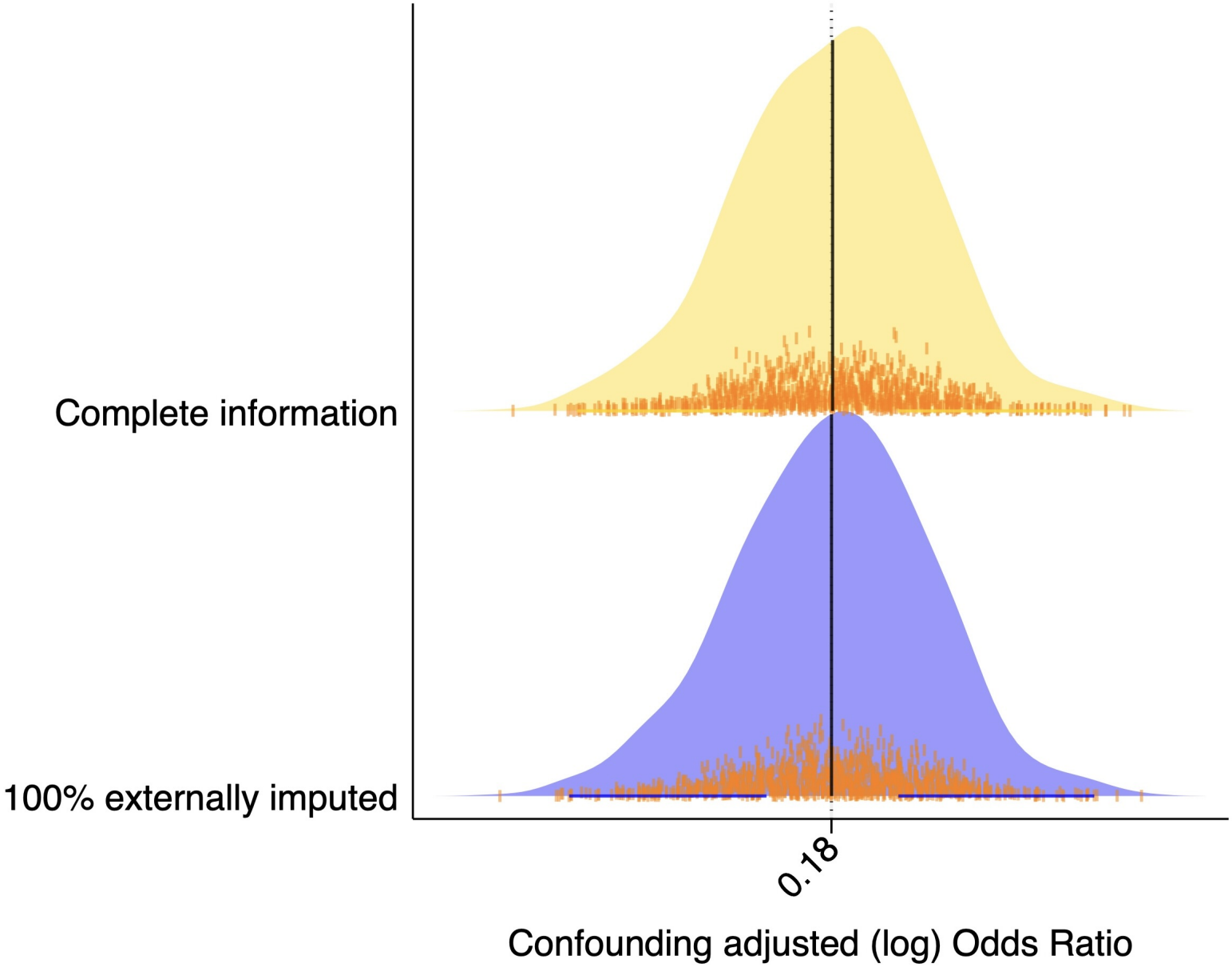
Beta-distribution



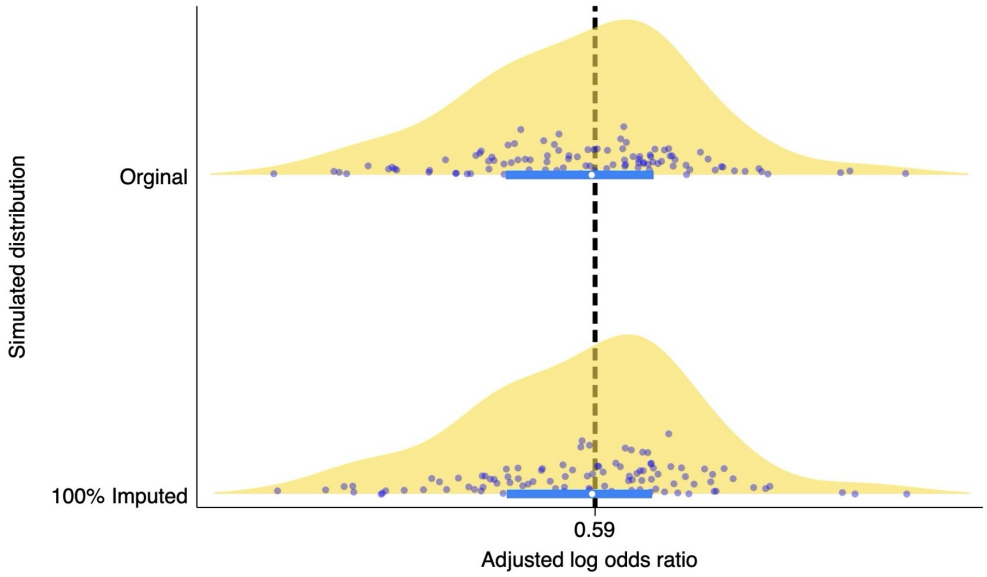
Simulation study

- Evaluation of the quantile imputation with external data in a variety of settings.
- The variable that was set to systematically missing in some studies was a strongly right-skewed variable generated from a χ^2-distribution with varying degrees of freedom.
- The target of statistical inference was the confounding-adjusted odds ratio.

Main results: No substantial bias larger than 1.5%



Magnitude of confounding between studies is very different



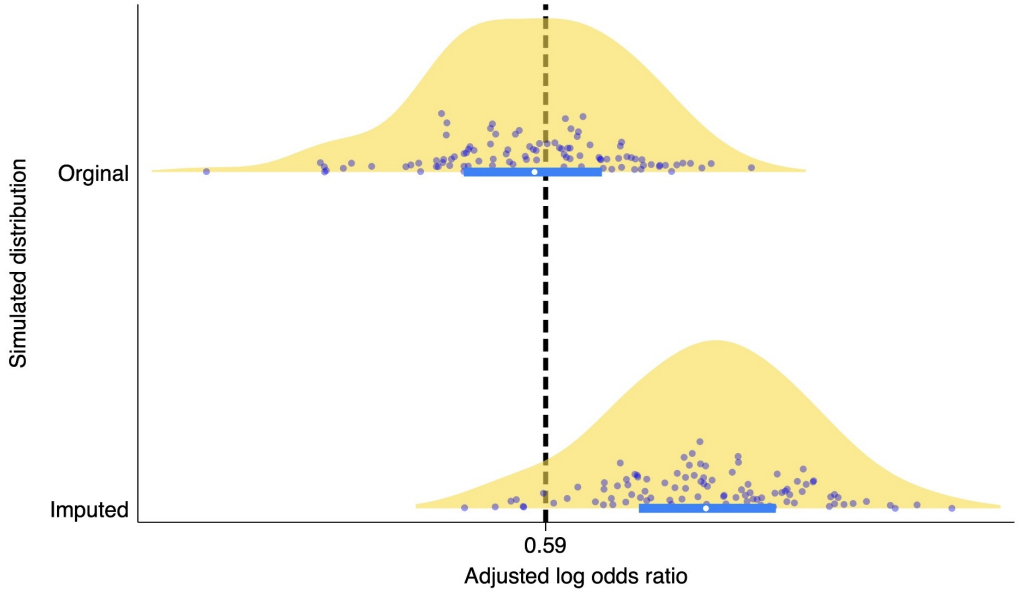


Table with the imputed estimates

	Study A N=4,182	**Study B** N=2,260	**Study C** N=1,401	**Study D** N=3,766	**Study E** N=2,382
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Conclusion

How do we impute missing data from external data sources?

- Conventional multiple imputation approaches are infeasible without individual-level data.
- We propose to fit a prediction model in an external study and share the regression coefficients to generate imputations.
- Quantile imputation is a flexible imputation method for continious variables.
- Next steps: comparison with other methods and evaluation of more complex scenarios.

Additional information

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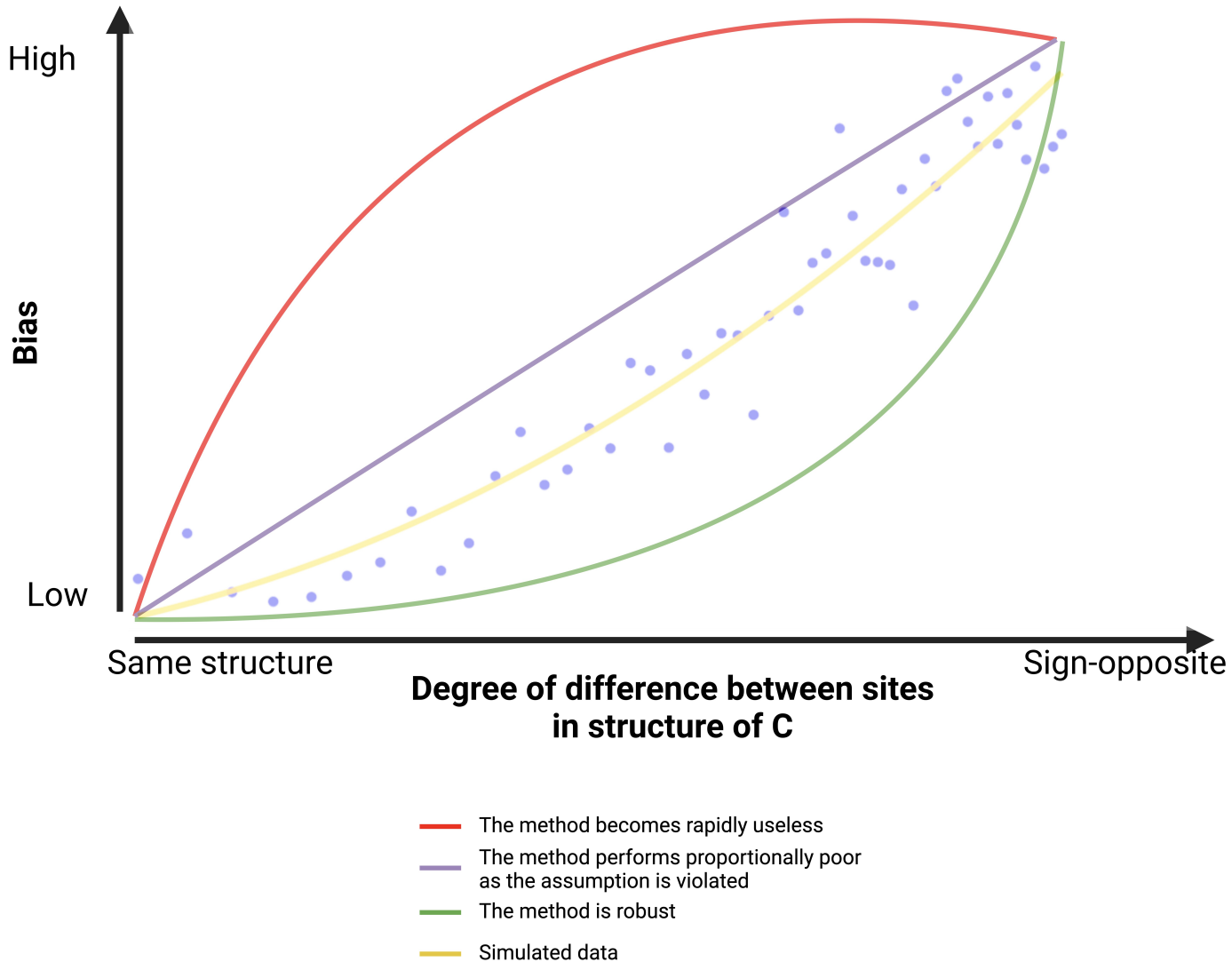
In collaboration with Nicola Orsini, Matteo Bottai, Viktor Ahlqvist, and Paul Madley-Dowd

1. Thiesmeier, R., Bottai, M., Orsini, N. (2024). Systematically missing data in distributed data networks: multiple imputation when data cannot be pooled. *J. Stat. Comput. Simul.*, 94(17), 3807–3825.
2. Thiesmeier R, Bottai M, Orsini N. Imputing missing data with external data. arXiv:2410.02982
3. Thiesmeier R, Madley-Dowd P, Orsini N, Ahlqvist V. Cross-site multiple imputation for recovering variables without individual-level pooled data. *Under review*.

Assumptions and considerations

- Transportability assumption: the association between the auxiliaries and the imputation target are transferable across sites.
- Measurement assumption: the variable we predict with observed data *measures* the same concept of the target variable we wish to impute.
- How to capture the heterogeneity between different sites? Single vs. multiple studies as the basis of imputation.

Transportability assumption



Steps involved in imputing without data

1. Identify study site(s) with observed data on y_{ij}
2. For each study site j with observed data on y_{ij}, we fit a prediction model on y_{ij} conditional on $\mathbf{w}_{ij}$
   - **Option 1**: Select single study as the basis for imputation - get regression coefficients: $\hat{\gamma}_j$.
   - **Option 2**: Select multiple studies as the basis for the imputation - get a weighted average of the regression coefficients $\bar{\hat{\gamma}}_j$
3. Transfer regression coefficients to sites with systematically missing values on y_{ij}.
4. Impute y_{ij} using $\hat{\gamma}_j$ or $\bar{\hat{\gamma}}_j$ and $\mathbf{w}_{ij}$ M-times.